

STAT 112

INTRODUCTION TO STATISTICS AND PROBABILITY II

SESSION 7- Poisson Random Variable & Distribution

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Session Outline

- 1 Poisson Process
- 2 Poisson Distribution
- 3 Moments
- 4 Practical Examples

Session Overview

In this session, students are introduced to Poisson probability distribution. The experiment that gives rise to the Poisson random variable and the properties of the Poisson distribution will be discussed. Computation of the probabilities using the Poisson probability mass function and theoretical derivation of the mean and variance of the Poisson random variable will be presented.

Learning Outcomes

At the end of the session, students will be able to:

- 1 Describe and explain the Poisson process.
- 2 State the probability mass function of the Poisson distribution and show that it is a probability function.
- 3 Prove the expectation and variance of Poisson distribution.
- 4 Compute probabilities and moments using the Poisson distribution function.

The Poisson Process

- The Poisson process which is the continuous analogue of the Bernoulli process is defined in terms of the probabilistic of the behaviour of the arrival of points on a continuous real line.
- Consider the time interval from $(0, t]$ and let the random variable X be the total number of events occurring in this interval.
- Suppose also that, we sub-divide this interval into n , non-overlapping sub-intervals of length $\frac{t}{n}$ and let $x_1, x_2, \dots, x_n$, respectively be the number of events occurring in each sub-interval.

Definition of Poisson Process

- A process is said to be Poisson if the following postulates are appropriate or they apply:
 - 1 The occurrence of an event in any given interval is independent of any other interval.
 - 2 The probability of a single occurrence of an event in a given interval is proportional to the length of the interval.
 - 3 In any infinitesimally small portion of an interval, the probability of more than one occurrence of an event is negligible. That is, in each small interval, the occurrence may either one or zero.

Poisson Distribution

Let X be a discrete random variable assuming the possible values $0, 1, 2, \dots$.

Then the Poisson probability distribution is defined for the number of events x , occurring in a fixed interval of time or space, if these events happen at a known average rate, λ and independently of the time as:

Poisson Probability Mass Function

$$P(X = x) = \frac{\lambda^x e^{-\lambda}}{x!}, \quad \text{where } x = 0, 1, 2, \dots$$

We say that, the random variable X has a Poisson distribution with parameter $\lambda > 0$ and λ is known as the Poisson rate.

Legitimacy Condition

First we review the MaClaurin expansion of the function e^x :

Proof of Legitimacy:

$$e^x = \sum_{r=0}^{\infty} \frac{x^r}{r!}, \quad r = 0, 1, 2, \dots$$

1 It is clear that, $P(X = x) = \frac{\lambda^x e^{-\lambda}}{x!} > 0, \forall x$, since $\lambda > 0$, $x! > 0$ and $e^{-\lambda} > 0$.

2

$$\begin{aligned} \sum_{all\ x} P(X = x) &= \sum_{x=0}^{\infty} \frac{\lambda^x e^{-\lambda}}{x!} \\ &= e^{-\lambda} \sum_{x=0}^{\infty} \frac{\lambda^x}{x!} \\ &= e^{-\lambda} e^{\lambda} = e^0 = 1 \end{aligned}$$

Moments

Theorem 1:

If X is random variable having a Poisson distribution with parameter λ , then

$$E(X) = \lambda \quad \text{and} \quad \text{Var}(X) = \lambda.$$

Proof:

By definition the expected value of X is given by:

$$\begin{aligned} E(X) &= \sum_{\text{all } x} xP(X = x) = \sum_{x=0}^{\infty} x \frac{\lambda^x e^{-\lambda}}{x!} \\ &= e^{-\lambda} \sum_{x=0}^{\infty} x \frac{\lambda^x}{x(x-1)!} \\ &= e^{-\lambda} \sum_{x=1}^{\infty} \frac{\lambda^x}{(x-1)!} \end{aligned}$$

Moments

Now we apply change of variable by letting $x = y + 1$, then when $x = 1$, $y = 0$, we have:

Proof Cont'd:

$$\begin{aligned} E(X) &= e^{-\lambda} \sum_{y=0}^{\infty} \frac{\lambda^{y+1}}{y!} \\ &= e^{-\lambda} \sum_{y=0}^{\infty} \lambda \frac{\lambda^y}{y!} = \lambda e^{-\lambda} \sum_{y=0}^{\infty} \frac{\lambda^y}{y!} \\ &= \lambda e^{-\lambda} e^{\lambda} \\ &= \lambda \end{aligned}$$

Proof of Variance:

By definition the variance of X is given by:

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = E(X^2) - \lambda^2$$

$$E(X^2) = \sum_{\text{all } x} x^2 P(X = x) = \sum_{x=0}^{\infty} x^2 \frac{\lambda^x e^{-\lambda}}{x!}$$

$$= e^{-\lambda} \sum_{x=0}^{\infty} [x(x-1) + x] \frac{\lambda^x}{x!}$$

$$= e^{-\lambda} \sum_{x=0}^{\infty} x(x-1) \frac{\lambda^x}{x(x-1)(x-2)!} + e^{-\lambda} \sum_{x=0}^{\infty} x \frac{\lambda^x}{x!}$$

$$= e^{-\lambda} \sum_{x=2}^{\infty} \frac{\lambda^x}{(x-2)!} + E(x)$$

Moments

Now we apply change of variable by letting $x = y + 2$, then when $x = 2$, $y = 0$ and we have:

Proof Cont'd:

$$\begin{aligned} E(X^2) &= e^{-\lambda} \sum_{y=0}^{\infty} \frac{\lambda^{y+2}}{y!} + \lambda \\ &= e^{-\lambda} \lambda^2 \sum_{y=0}^{\infty} \frac{\lambda^y}{y!} + \lambda \\ &= \lambda^2 e^{-\lambda} e^{\lambda} + \lambda = \lambda^2 + \lambda \end{aligned}$$

$$\begin{aligned} \therefore \text{Var}(X) &= \lambda^2 + \lambda - \lambda^2 = \lambda \\ &= \lambda \end{aligned}$$

Example 1:

Suppose that in an email management company, an email user gets, on average, 147 emails per day. If the number of emails received per day follows a Poisson distribution with mean 147 emails, compute the following:

- a What is the probability that an email user receives exactly 160 emails per day?
- b What is the probability that an email user receives at most 160 emails per day?
- c What is the standard deviation of the number of emails received per day?

Solution:

Let X denote the number of emails received per day, this means that x can assume 0, 1, 2, 3, ... and $X \sim P(147)$.

- a The probability that exactly 160 emails user received is given by:

$$P(X = 160) = \frac{\lambda^x e^{-\lambda}}{x!} = \frac{(147)^{160} e^{-147}}{160!}$$
$$\approx 0.018.$$

- b The probability that an email user receives at most 160 emails per day is given as:

$$P(X \leq 160) = \sum_{x=0}^{160} \frac{(147)^x e^{-147}}{x!}$$
$$\approx 0.8666.$$

- c Standard deviation $\sqrt{\text{Var}(X)} = \sqrt{147} = 12.1244.$

Practical Examples

Example 2:

Suppose that Gabby's answering machine receives about six telephone calls between 8:00 a.m. and 10:00 a.m.

What is the probability Gabby receives more than one call in the next 15 minutes?

Solution:

Let X be the number of calls Gabby receives in 15 minutes.

Note; the interval of interest is 15 minutes or $\frac{1}{4}$ hour.

The random variable X takes values $0, 1, 2, 3, \dots$

Solution:

Solution Cont'd

If Gabby receives, on average, six telephone calls in two hours, and there are eight 15-minute intervals in two hours, then Gabby receives, $\lambda t = 6 \times \frac{1}{8} = 0.75$ calls in 15 minutes, on average.

$\Rightarrow X \sim P(0.75)$ and the probability that Gabby receives more than one call in the next 15 minutes is given as:

$$P(X > 1) = 1 - P(X \leq 1) = 1 - \sum_{x=0}^1 \frac{(0.75)^x e^{-0.75}}{x!} = 0.1734.$$

Trial Questions

- 1 The average number of loaves of bread put on a shelf in a bakery in half-hour period is 12. The baker is interested in the number of loaves of bread put on the shelf in five minutes interval. What is the probability that the number of loaves, selected randomly, put on the shelf in five minutes is three?
- 2 If 3% of the electric bulbs manufactured by a company are defective, find the probability that in a sample of 100 bulbs,
- 1 0,
 - 2 1,
 - 3 3,
 - 4 4,
 - 5 5,
- bulbs will be defective.

THANK YOU!!!